ECE 20007 Audio Equalizer Final Project Report

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Abstract—

This project presents the design, construction, and experimental evaluation of an audio equalizer incorporating RC filters, operational amplifiers, and a power amplifier. The circuit divides an input audio signal into low-, mid-, and high-frequency bands using dedicated filters, each with an independently adjustable gain. The filtered signals are recombined through a summing amplifier that provides overall volume control, after which the signal is amplified by a power amplifier to drive a speaker. Frequency response analysis and output voltage measurements verified that system performance remained within acceptable error limits. The completed design demonstrated effective frequency shaping, adequate power delivery, and clear audio playback.

Keywords—audio equalizer, filters, op amps, power amplifier

I. INTRODUCTION

A. Definition and Purpose

The objective of this project is to design, construct, and experimentally validate the operation of an audio equalizer. This is achieved through the use of frequency-selective filtering, operational amplifiers, buffering stages, signal recombination, and power amplification. Together, these components enable an input audio signal to be separated into low-, mid-, and high-frequency bands, each with an independently adjustable gain. The processed signals are then recombined and amplified to drive a speaker.

This project reinforces the electrical circuit concepts studied throughout the course by applying them in a practical, system-level design. By integrating multiple circuit elements into a cohesive design, the project culminates in the successful construction of a functional audio equalizer capable of clearly reproducing audio signals.

The specific objectives of this project are as follows:

- Design an audio equalizer that meets all specified performance requirements.

- Develop circuit schematics and perform simulations to verify functionality.
- Construct and evaluate the low-pass, band-pass, and high-pass filter stages.
- Design and test the volume control and summing amplifier circuits.
- Integrate the power amplifier and evaluate the overall circuit output.
- Connect an audio source and speaker to assess full system performance.

B. Key Functions

The primary purpose of an audio equalizer is to shape the frequency content of an input audio signal by selectively amplifying or attenuating specific frequency ranges. This is achieved by first splitting the input signal into low-pass, band-pass, and high-pass paths using frequency-selective filters. Each path includes an independent volume control, allowing the user to adjust the amplitude of each frequency band and thereby tailor the overall spectral balance of the output signal.

The individually adjusted signals are then recombined using a summing amplifier, which also provides a master volume control. The combined signal is subsequently fed into a power amplifier that increases the signal strength and drives the speaker. Together, these stages enable precise control over the frequency balance of the audio output while maintaining sufficient output power and uninterrupted speaker operation.

C. Applications

Audio equalizers are widely utilized in both consumer and professional audio systems to provide flexible control over sound characteristics. By allowing selective amplification or attenuation of specific frequency ranges, equalizers enable users to tailor audio output to match personal listening preferences, compensate for speaker limitations, or adapt to environmental constraints. In consumer electronics, equalizers are commonly integrated into smartphones, automotive sound systems, headphones, and Bluetooth speakers, where they enhance perceived audio quality despite size, power, or hardware limitations.

In professional applications, audio equalizers play a critical role in recording studios, broadcast environments, and live sound systems. They are used to correct for room acoustics, reduce unwanted resonances or feedback, and ensure consistent sound reproduction across different listening spaces. By adjusting frequency response to suit the acoustic properties of a venue, equalizers help maintain clarity, balance, and intelligibility in both recorded and live audio, making them an essential component of modern audio signal processing systems.

II. THEORY

A. General Filters Working Principle

In this project, low-pass, band-pass, and high-pass filters were used to shape the frequency content of the final output signal. These filters rely on the frequency-dependent behavior of capacitors to regulate how much of the input signal reaches the output. In alternating current circuits, capacitors do not exhibit a fixed resistance; instead, they have a reactance that varies with frequency. At low frequencies, a capacitor behaves similarly to an open circuit, while at higher frequencies it approaches the behavior of a short circuit. By selecting appropriate resistor and capacitor values and adjusting the placement of these components within a circuit, filters can be designed to selectively pass or attenuate specific frequency ranges.

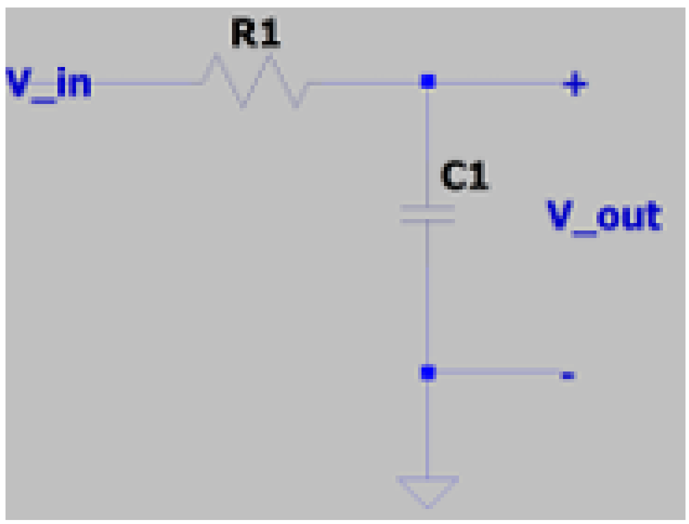

Fig. 1. RC lowpass filter circuit diagram

In an RC low-pass filter, the resistor is placed in series with the input signal, while the capacitor is connected to ground. At low frequencies, the capacitor's reactance is large, causing minimal current to flow through it and allowing most of the input voltage to appear at the output. As the frequency increases, the reactance of the capacitor decreases, directing more of the signal to ground and reducing the output voltage. This behavior causes higher frequencies to be attenuated while lower frequencies are preserved. The cutoff frequency is defined as the point at which the output amplitude begins to decrease significantly and can be calculated using Equation (1). Frequencies below this cutoff generally pass through the filter, while frequencies above it are heavily attenuated.

$$f_c = \frac{1}{2\pi R_1 C_1} \quad (1)$$

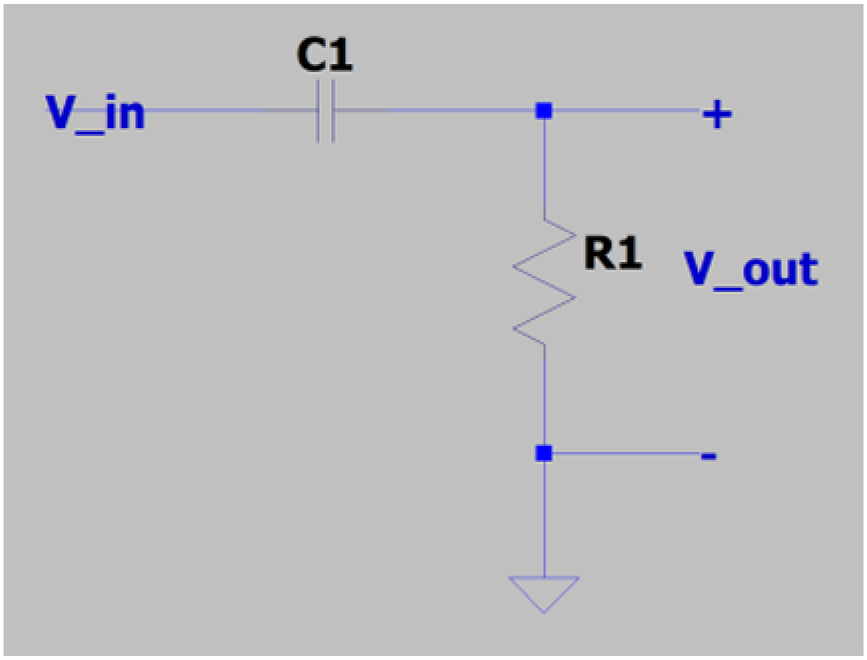

Fig. 2. RC high pass filter circuit diagram

An RC high-pass filter operates on the same principles as the low-pass filter but with the component positions reversed. In this configuration, the capacitor is placed in series with the input signal and the resistor is connected to ground. At low frequencies, the large reactance of the capacitor prevents most of the input signal from reaching the output. As the frequency increases and the capacitor's reactance decreases, more of the signal is able to pass through the capacitor and develop across the resistor. As a result, frequencies above the cutoff value are passed, while lower frequencies are attenuated. The cutoff frequency for the high-pass filter is also determined using Equation (1).

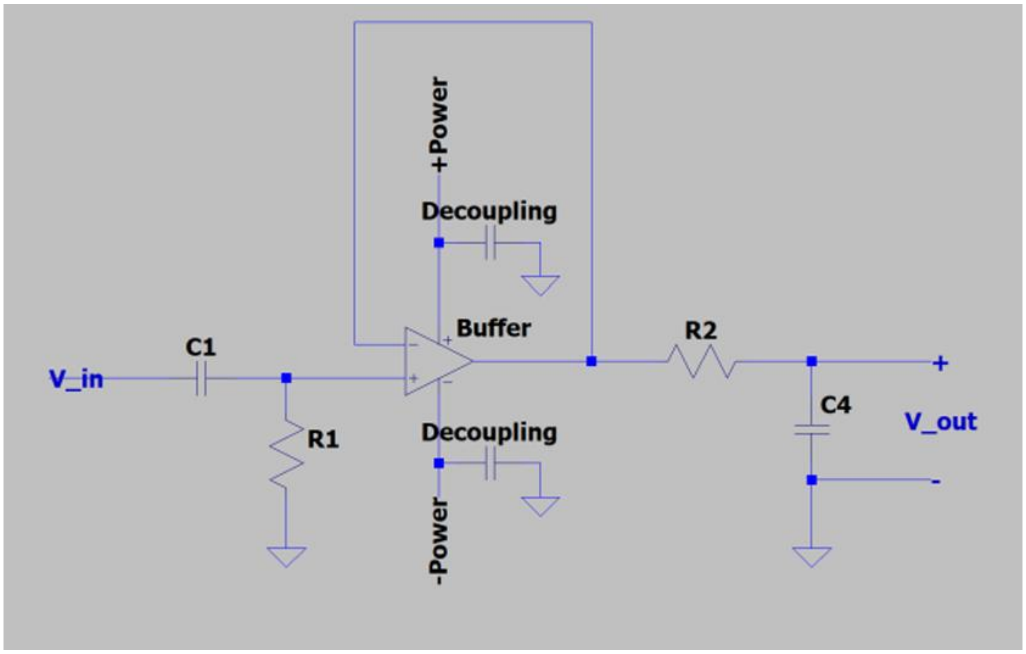

Fig. 3. RC mid-pass filter circuit diagram

In the final circuit design, the band-pass filter was implemented by cascading an RC high-pass filter with an RC low-pass filter. Each stage operates independently as previously described. The high-pass filter first removes frequency components below its cutoff frequency, allowing higher-frequency signals to pass to the subsequent low-pass filter. The low-pass filter then attenuates frequencies above its cutoff, producing an output that contains only frequencies within a defined range. Together, these filters create a passband bounded by the cutoff frequency of the high-pass filter and that of the low-pass filter. These cutoff frequencies are calculated using Equation (1) and the selected resistor and capacitor values.

To ensure proper filter performance, a buffer stage is inserted between the high-pass and low-pass filters. This buffer isolates the two filter stages and prevents loading effects that could otherwise alter the intended cutoff frequencies. The operating principles and theoretical justification for the buffer will be discussed in the following section.

B. Operational Amplifiers Working Principle

Operational amplifiers were used throughout the audio equalizer circuit to perform a variety of essential functions. An operational amplifier produces an output voltage proportional to the difference between its two input terminals. Due to the very high input impedance of these terminals, minimal current is drawn from connected circuit elements, allowing the surrounding circuitry to behave predictably and enabling op amps to be configured for a wide range of applications.

In addition, operational amplifiers require a stable power supply to operate correctly. To maintain supply stability and reduce noise, decoupling capacitors were incorporated into the circuit. These capacitors help suppress voltage fluctuations and ensure reliable op amp performance across all stages of the system.

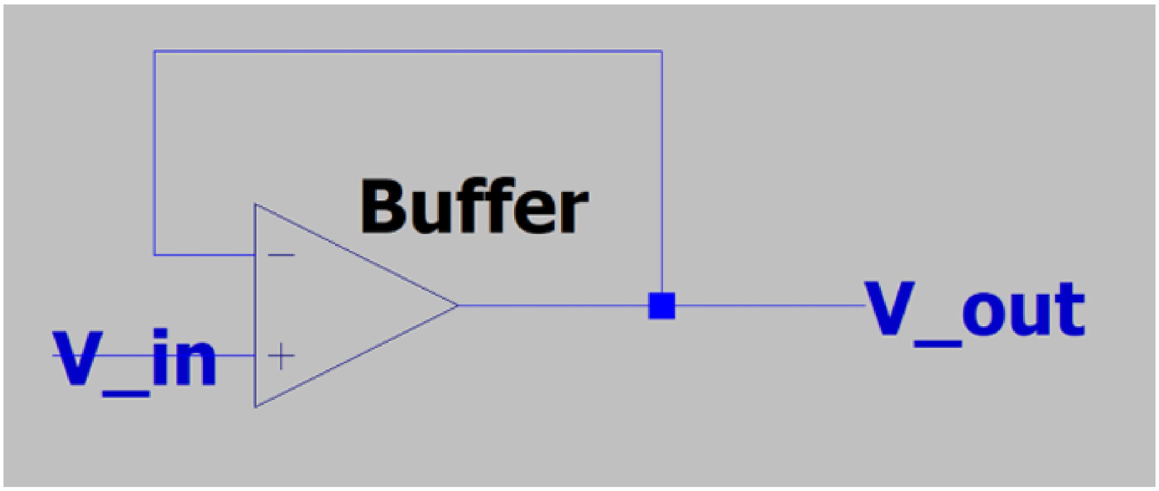

Fig. 4. Buffer circuit diagram

As mentioned previously, we used a buffer in our circuit to create the mid-pass filter, placing it between the lowpass and high pass filters. The buffer op amp connects the output directly to the inverting input, which makes the output voltage equal to the input voltage through negative feedback. Within the op amp, the inputs have extremely high resistance between them, meaning that the buffer essentially creates an open circuit. This separates the input side of the filter from the output, which prevents loading between the filter stages. Without the buffer, the impedance of one stage of the filter would influence the behavior of the next stage, which would affect the cutoff frequencies. Ultimately, this would cause the pass band of the mid-pass filter to deviate from our calculations.

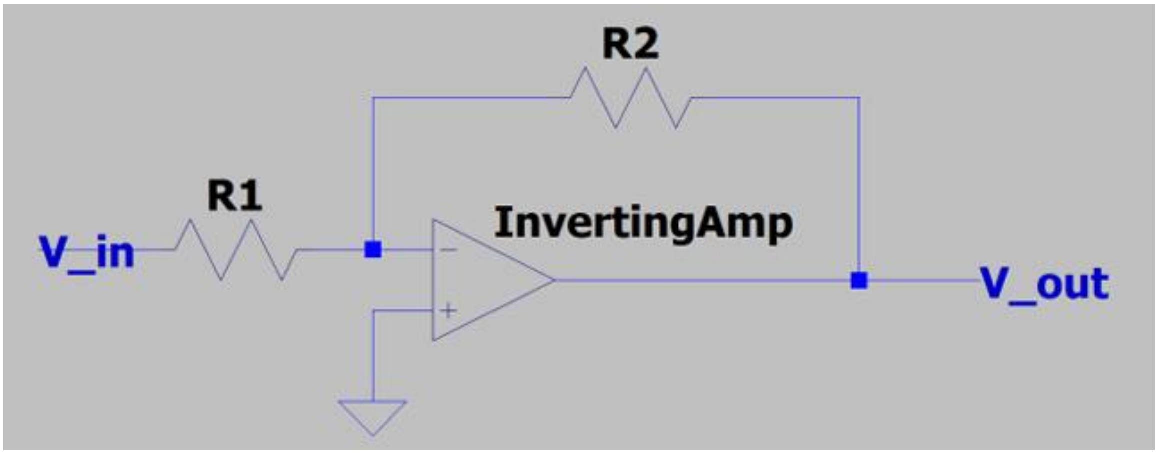

Fig. 5. Inverting amplifier circuit diagram

In the final audio equalizer design, inverting amplifier configurations were employed to implement volume control within the low-pass, band-pass, and high-pass signal paths. In this configuration, the input signal is applied through a resistor to the inverting terminal of the operational amplifier, while the non-inverting terminal is connected to ground. This arrangement places the op amp in a negative feedback configuration, as the output is fed back to the inverting input through a second resistor.

Negative feedback forces the op amp to adjust its output such that the voltage at the inverting terminal remains close to zero, often referred to as a virtual ground. Because the op amp draws negligible input current, the current flowing through the input resistor R_1 must pass through the feedback resistor R_2. This relationship directly determines the ratio of the output voltage to the input voltage, commonly referred to as the amplifier gain, which can be calculated using Equation (2).

To provide adjustable volume control, a potentiometer was used as the feedback resistor R_2, resulting in an inverting amplifier with variable gain. This design allows the user to easily and precisely adjust the amplitude of each frequency band before signal recombination.

$$\frac{v_{\text{out}}}{v_{\text{in}}} = -\frac{R_2}{R_1} \qquad (2)$$

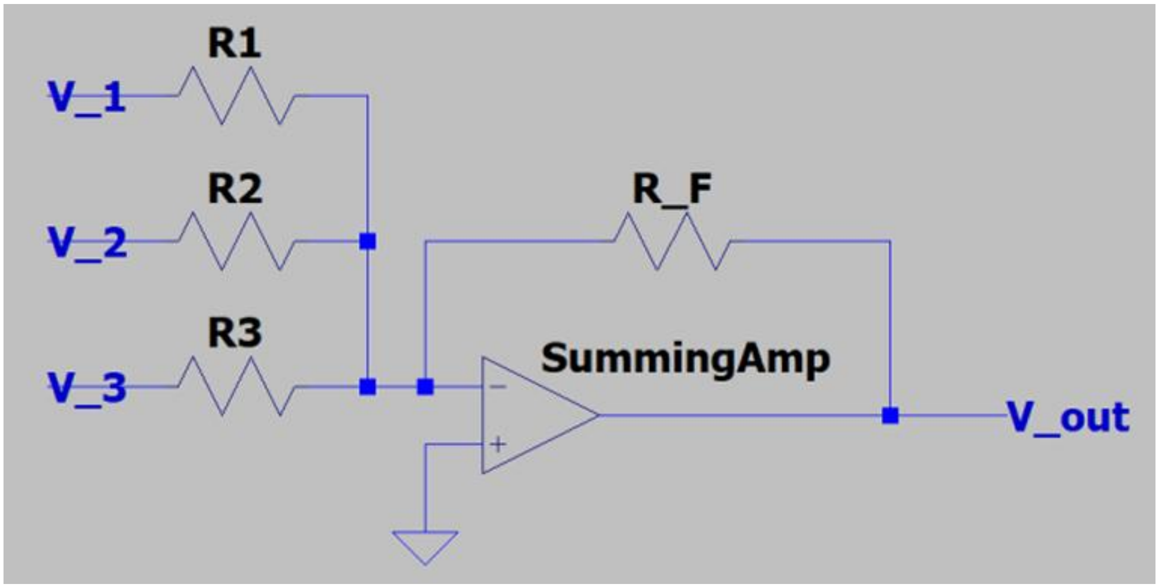

Fig. 6. Summing amplifier circuit diagrams

$$V_{\text{out}} = -R_F\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right) \quad (3)$$

$$R_{\text{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)^{-1} \quad (4)$$

The final circuit employs a summing amplifier to recombine the signals from the low-pass, band-pass, and high-pass filter branches. A summing amplifier combines multiple input signals into a single output by applying each signal, through its respective resistor, to the inverting terminal of an operational amplifier. The resulting input currents are summed at the inverting node before entering the op amp.

Due to the high input impedance of the operational amplifier, these summed currents are forced to flow through the feedback resistor. This configuration produces an output voltage equal to the inverted, weighted sum of the individual input voltages, as described by Equation (3). In this design, no individual frequency band was intended to have greater influence than the others, so equal resistor values were used for each input branch. Equation (4) was then applied to determine the equivalent resistance of the summing network, which was used to calculate the overall gain of the summing amplifier stage.

C. Power Amplifiers Working Principle

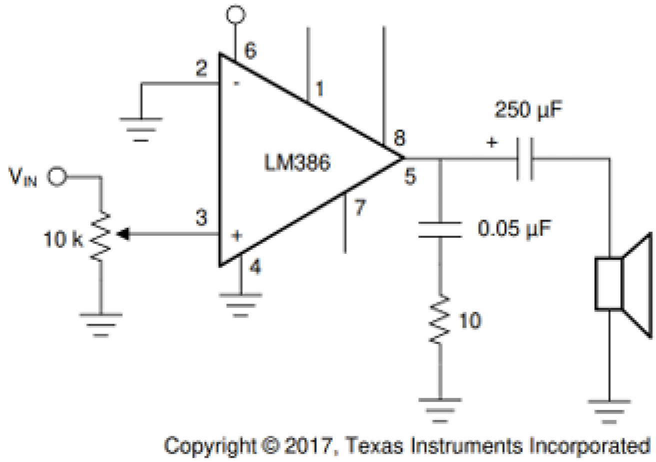

Fig. 7. Power amplifier (LM386) circuit diagram

The power amplifier represents the final stage of the audio equalizer and serves to amplify the signal produced by the summing amplifier before it is delivered to the speaker. Unlike standard operational amplifiers, power amplifiers are specifically designed to provide substantially higher output current, enabling them to drive low-impedance loads effectively. In this system, the power amplifier increases the current capability of the low-power audio signal from the summing amplifier while preserving the original waveform.

Whereas the operational amplifiers used in earlier stages primarily facilitate voltage amplification and signal conditioning, the power amplifier is optimized to supply the large currents required by low-impedance devices such as the 8 Ω speaker used in this project. The LM386 power amplifier was selected for this application due to its class AB configuration, which offers efficient operation and low distortion while maintaining similarities to standard op-amp behavior.

In the final circuit configuration, shown in Figure 7, the LM386 is operated in its standard non-inverting amplifier mode. The input signal is applied directly to the non-inverting terminal and amplified internally. Additional resistors and capacitors are incorporated to block DC offset components and reduce noise, ensuring stable and reliable audio output.

III. PROCEDURE

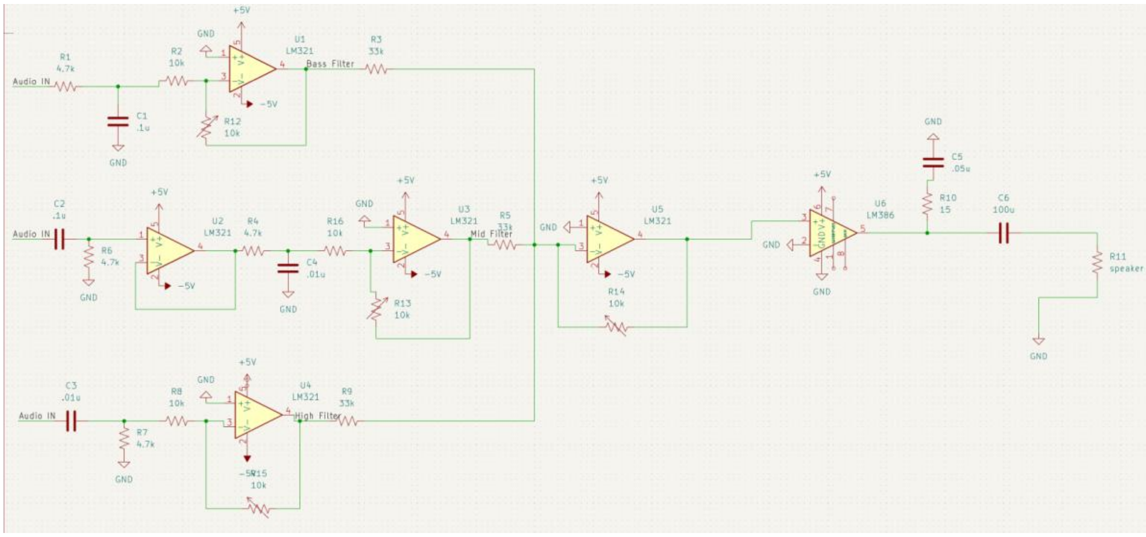

Fig. 8. Schematic of the audio equalizer

A. Design Parameters

1) *Filters:* For the low-pass filter, the desired cutoff frequency was set to 320 Hz. Since a wider selection of resistor values was available, the capacitance was chosen first, with a value of 0.1 μF selected for the design. Equation (1) was then used to determine the required resistance. Rearranging the equation to solve for R_1 yields:

$$f_c = \frac{1}{2\pi R_1 C_1} \qquad (5)$$

$$R_1 = \frac{1}{2\pi(0.1 \times 10^{-6})(320)} = 4974 \approx 5000\ \Omega$$

As shown above, substituting the selected capacitance and cutoff frequency into the design equation yields a required resistance of approximately 5 kΩ. Using the available components in our resistor kit, this value was closely approximated by placing a 4.7 kΩ resistor in series with a 330 Ω resistor.

For the high-pass filter, the desired cutoff frequency was set to 3200 Hz. As with the low-pass filter, a 0.1 μF capacitor was selected. Since the cutoff frequency expression for RC low-pass and high-pass filters is identical, the same calculation procedure was used, with the cutoff frequency adjusted from 320 Hz to 3200 Hz.

$$R_1 = \frac{1}{2\pi(0.1 \times 10^{-6})(3200)} = 497.4 \approx 500\ \Omega$$

As shown, these values yield a resistance of approximately 500 Ω. In the final circuit implementation, this resistance was closely approximated by connecting a 470 Ω resistor in series with a 33 Ω resistor.

The band-pass filter in the final design was implemented by cascading a high-pass filter, a buffer stage, and a low-pass filter. To determine the appropriate resistor and capacitor values for the high-pass and low-pass sections, Equations (1) and (7) were used. For these stages, 0.01 μF capacitors were selected. The high-pass filter cutoff frequency was set to 320 Hz, while the low-pass filter cutoff frequency was set to 3200 Hz. Substituting these values into the corresponding equations allowed the required resistance values to be calculated for each filter stage.

$$R_1 = \frac{1}{2\pi(0.01 \times 10^{-6})(320)} = 49736 \approx 50000\ \Omega$$

$$R_2 = \frac{1}{2\pi(0.01 \times 10^{-6})(3200)} = 4974 \approx 5000\ \Omega$$

Based on the completed calculations, the required resistance for the high-pass filter was approximately 50 kΩ, while the low-pass filter required a resistance of approximately 5 kΩ. In the final circuit implementation, these values were closely approximated using standard resistor combinations. A 47 kΩ resistor was placed in series with a 3.3 kΩ resistor to achieve the 50 kΩ target, while a 4.7 kΩ resistor was combined in series with a 330 Ω resistor to approximate the 5 kΩ value.

2) *Volume Control:* For the volume control stages in the low-pass, high-pass, and band-pass branches, the gain was designed to vary from 0 to −1. This range allows the user to selectively attenuate specific frequency bands by adjusting the corresponding potentiometers. To implement this functionality, inverting amplifier configurations were used, with the potentiometers placed at the feedback resistor position R_2. The gain, defined as the ratio of the output voltage to the input voltage, is determined by Equation (4).

$$\frac{v_{\text{out}}}{v_{in}} = -\frac{R_2}{R_1} = -\frac{0}{10000} = 0$$

$$\frac{v_{\text{out}}}{v_{\text{in}}} = -\frac{R_2}{R_1} = -\frac{10000}{10000} = -1$$

Using this gain expression, the minimum and maximum resistance values of the potentiometer were substituted into the equation. In this design, the potentiometer ranges from approximately 0 Ω at its minimum setting to 10 kΩ at its maximum. Substituting these limits shows that, to achieve a gain range from 0 to −1, a fixed 10 kΩ resistor must be used at R_1, as illustrated above. This configuration was implemented consistently across the low-pass, band-pass, and high-pass volume control stages

3) *Signal Recombination:* It is specified that when all potentiometers are set to their maximum resistance, the output of the summing amplifier should be 100 mV $_{\text{RMS}}$. The output voltage of the summing amplifier can be calculated using Equation (3) by substituting the potentiometer resistance for $\mathrm{R_F}$. By setting $\mathrm{R_1} = \mathrm{R_2} = \mathrm{R_3} = \mathrm{R}$, the expression simplifies, allowing the required resistance value Rto be solved directly.

$$R = -\frac{R_F}{v_{\text{out}}}(v_1 + v_2 + v_3) \quad (6)$$

The calculated expression can now be evaluated by substituting the appropriate circuit values with all potentiometers set to their maximum resistance. Under these conditions, the feedback resistance $R_F$is equal to 10 kΩ, and the desired output voltage $v_{\text{out}}$is 100 $\mathrm{mV_{RMS}}$. To solve for R, the sum of the input voltages to the summing amplifier must first be determined.

In this circuit, the low-pass branch processes input signals with frequencies below 320 Hz, the high-pass branch processes signals above 3200 Hz, and the band-pass branch processes frequencies between these limits. Because the frequency bands do not overlap, only one branch is assumed to be active at any given time. Additionally, with all potentiometers at their maximum settings, the gain of each volume control stage is −1. As a result, the combined input to the summing amplifier is approximately −320 mV, corresponding to the inverted input signal. Substituting these values into the expression allows the required value of $R$to be calculated.

$$R = -\frac{10000}{0.100}(-0.320) = 32000\ \Omega$$

As shown, this gives a resistor value of 32 kilohms. In our final circuit, we approximated this value using a 33 kilohm resistor. We are also given that, with the potentiometers at their minimum resistance, the output voltage should be no greater than 15 $\mathrm{mV_{RMS}}$. Upon completing equation (3) again but with a value of 0 ohms as RF, it is clear that the output voltage will be approximately 0 V. This relationship between the potentiometer and output voltage also demonstrates the other function of the summing amplifier, which is as the master volume control. By adjusting the potentiometer, users can change the output of the summing amplifier from practically 0 V to the 100 $\mathrm{mV_{RMS}}$ value. This ultimately changes the final volume of the speaker from the max to almost 0.

B. Operation of the Designed Circuit

The circuit operation begins with the input audio signal, which is divided into three parallel signal paths. The first path contains an RC low-pass filter that passes frequencies below 320 Hz while attenuating higher-frequency components. The second path includes an RC high-pass filter that passes frequencies above 3200 Hz and attenuates lower-frequency signals. The third path implements a band-pass filter constructed from a high-pass filter, a buffer stage, and a low-pass filter, allowing frequencies between 320 Hz and 3200 Hz to pass while suppressing frequencies outside this range.

Following frequency filtering, the signals from each branch are routed to individual volume control stages implemented with operational amplifiers and potentiometers. These stages allow independent adjustment of the amplitude of each frequency band before recombination. The filtered and adjusted signals are then combined using a summing amplifier, which includes an additional potentiometer to provide overall volume control. Finally, the combined signal is sent to the power amplifier, where the signal power is increased to drive the speaker.

IV. RESULTS

A. Filter Results

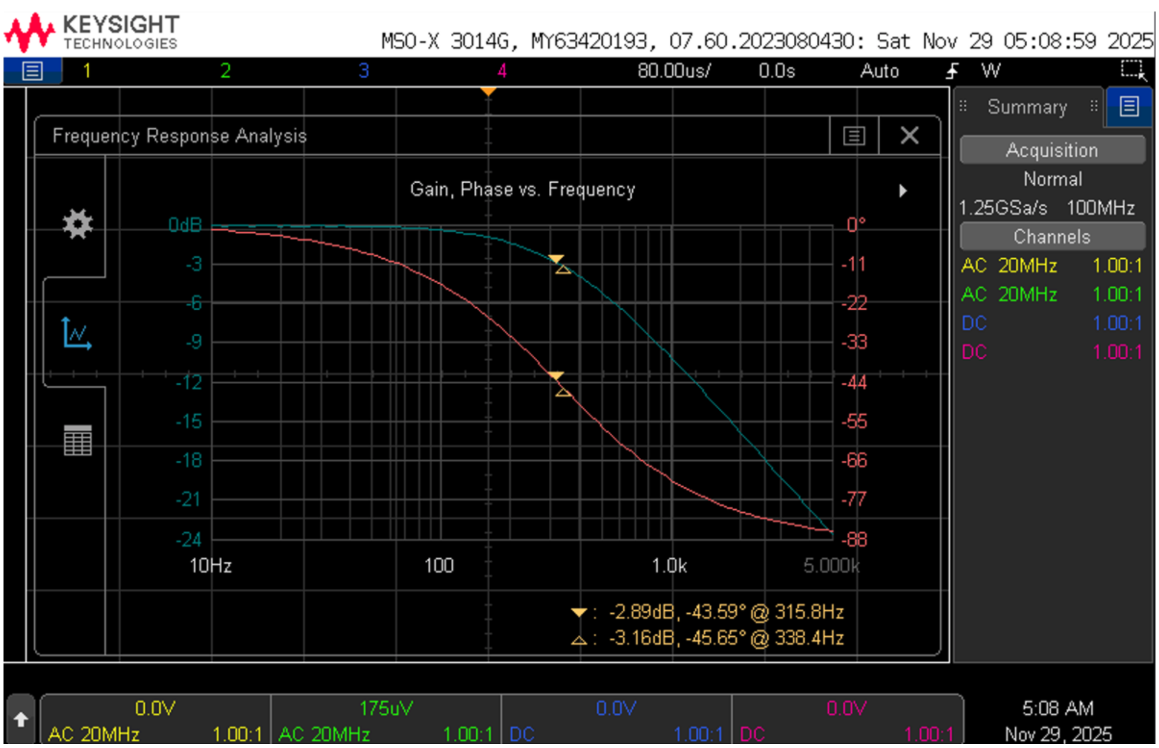

Fig. 9. Lowpass filter’s FRA curves

In the frequency response analysis shown above, the output of the low-pass filter was measured relative to the input voltage while sweeping the input frequency from 100 Hz to 10 kHz. As illustrated by the plot, the observed gain behavior closely matches the expected response of a low-pass filter. At lower frequencies, the signal is largely passed with minimal attenuation, whereas increasing attenuation is observed as the frequency rises. Using the measurement cursors, the −3 dB point was identified at approximately 306 Hz. This −3 dB point represents the frequency at which the output signal begins to be significantly attenuated and is therefore taken as the experimentally determined cutoff frequency of the low-pass filter.

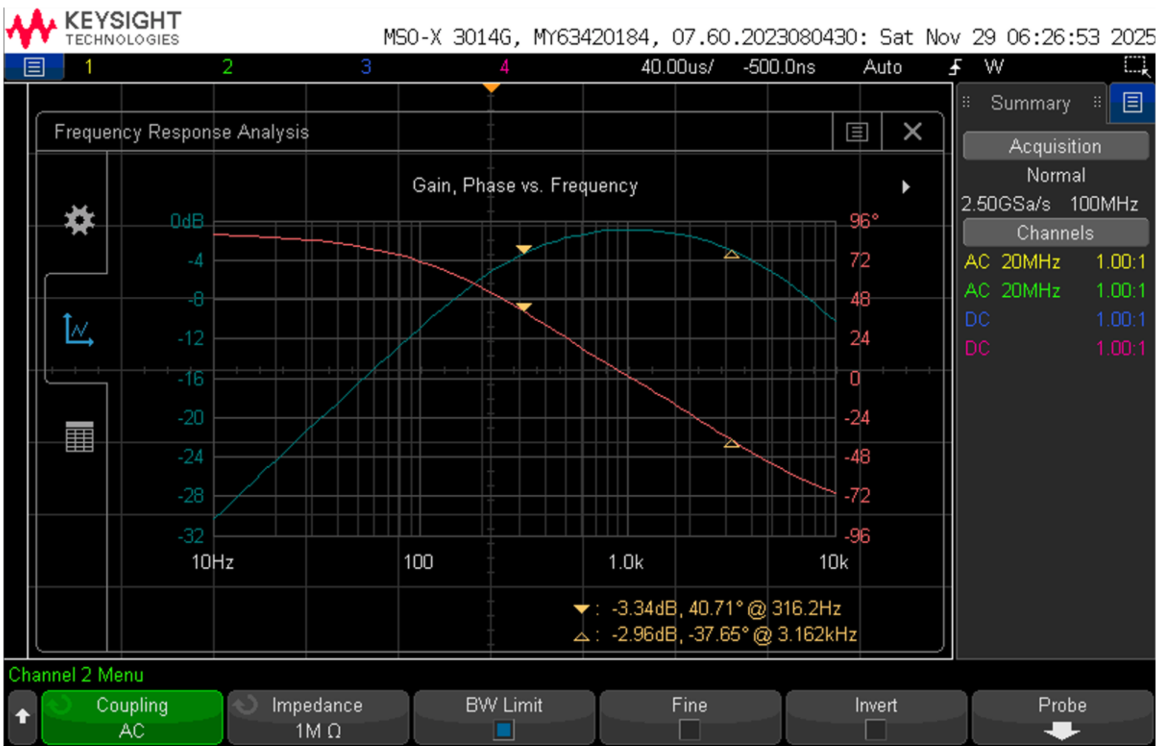

Fig. 10. Mid-pass filter's FRA curves

The frequency response analysis shown above presents the results for the band-pass filter, where the output voltage was measured relative to the input voltage over a frequency range of 100 Hz to 10 kHz. The resulting gain curve closely follows the expected behavior of a band-pass filter. At frequencies below the high-pass cutoff, the output signal is significantly attenuated. As the frequency increases beyond this cutoff, the signal begins to pass through the filter and the gain rises. This behavior continues until the frequency exceeds the cutoff frequency of the low-pass stage, at which point the output voltage decreases again.

Using the measurement cursors, the −3 dB points were identified at approximately 306 Hz and 3060 Hz. These values represent the experimentally determined lower and upper cutoff frequencies of the band-pass filter.

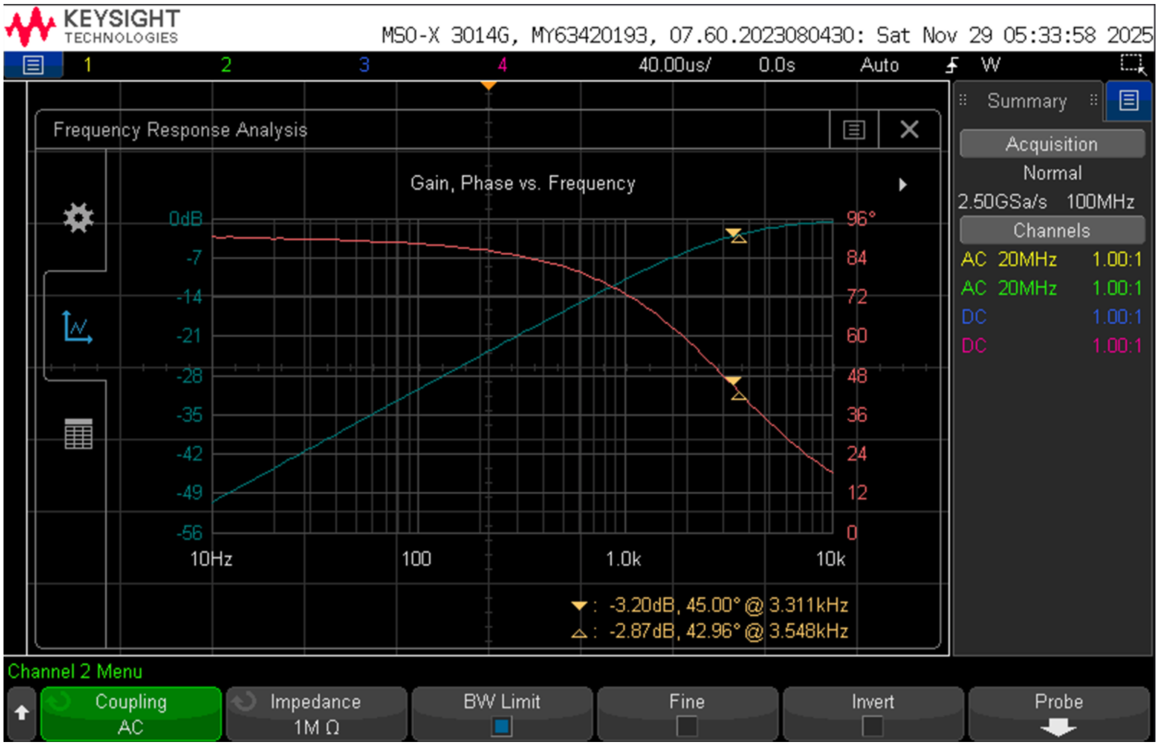

Fig. 11. High pass filter's FRA curves

The frequency response analysis shown above illustrates the performance of the high-pass filter and closely matches the expected theoretical behavior. At low frequencies, the output voltage remains small as most of the input signal is attenuated. As the frequency increases, a greater portion of the signal passes through the filter, continuing until the cutoff frequency is reached, beyond which the signal is largely passed with minimal attenuation. Using the measurement cursors, the −3 dB point was identified at approximately 3268 Hz, which is taken as the experimentally determined cutoff frequency of the high-pass filter.

B. V_{amp} Results at Minimum Settings

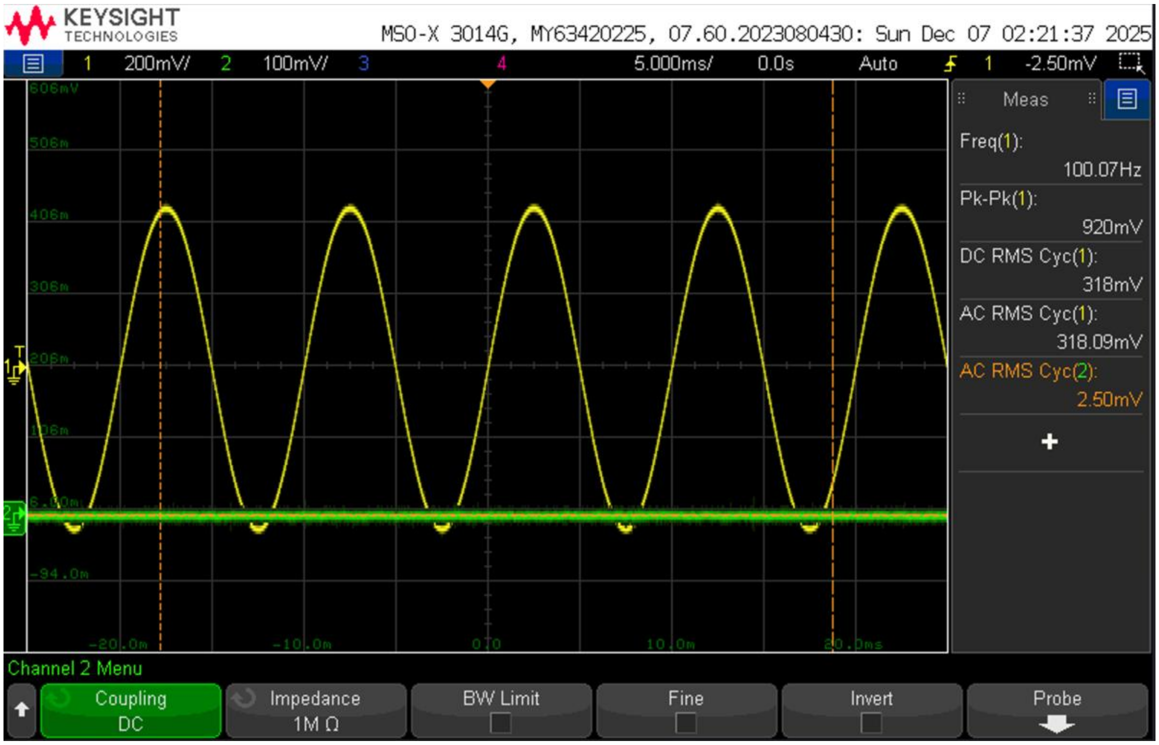

Fig. 12. V_{amp} at minimum settings for 100 Hz

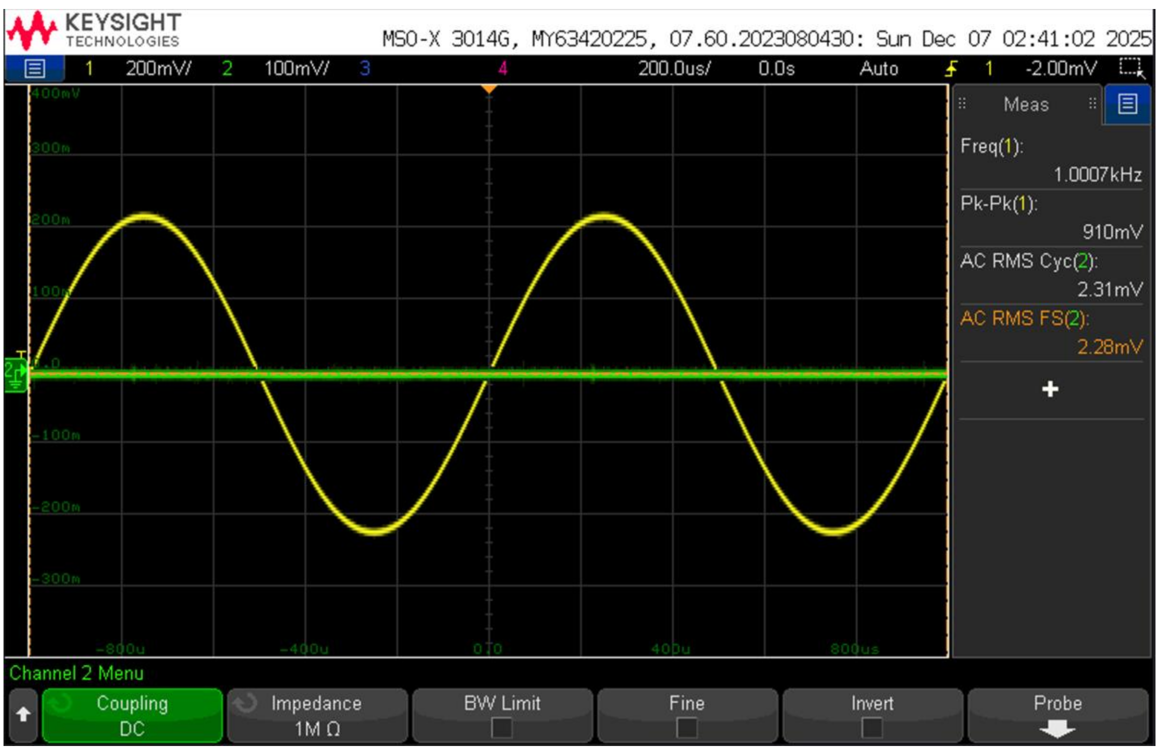

Fig. 13. V_{amp} at minimum settings for 1 kHz

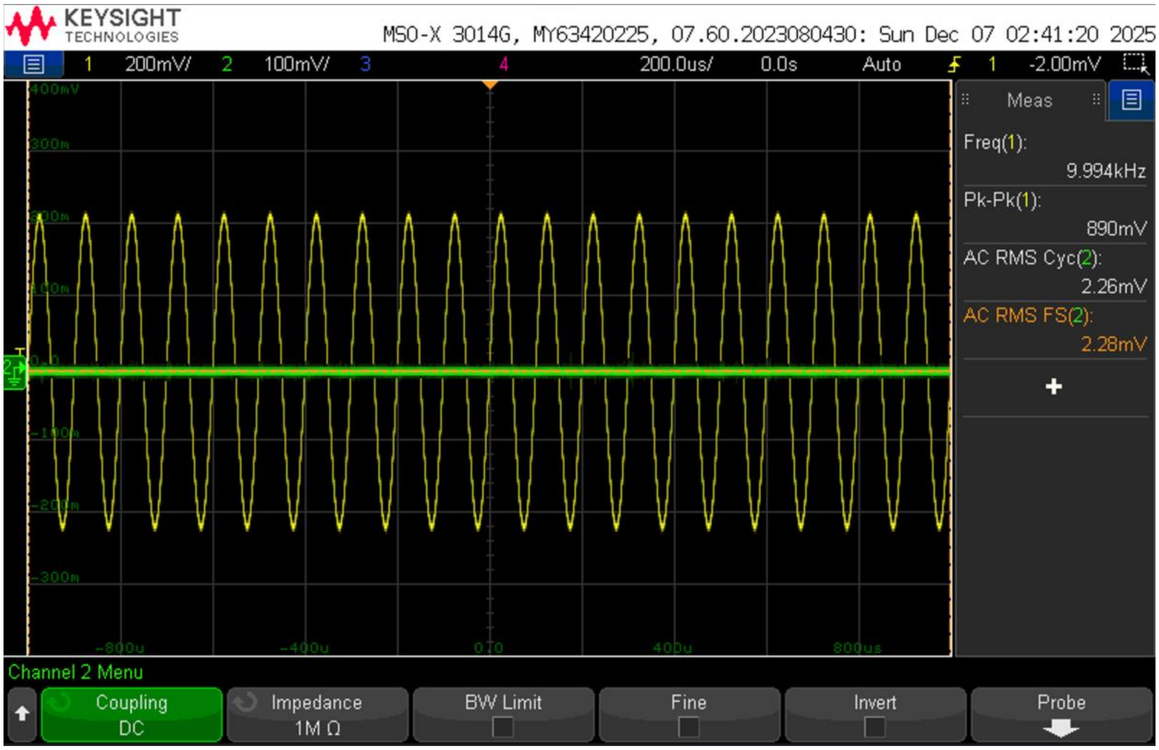

Fig. 14. V_{amp} at minimum settings for 10 kHz

The previous graphs can be summarized as follows: at an input frequency of 100 Hz, the measured output voltage of the summing amplifier was approximately 2.5 mV_{RMS}. At both 1 kHz and 10 kHz, the measured output voltage was approximately 2.28 mV_{RMS}. In all cases, the output voltage remained well below the specified maximum of 15 mV_{RMS}, indicating that the summing amplifier output meets the design requirements across the tested frequency range.

Clearly, when the volume control potentiometers are adjusted to their minimum settings, the output voltage is significantly reduced, which aligns with the expected behavior of the circuit. As discussed previously, the combination of potentiometers and resistors in the inverting op-amp volume control stages provides a variable gain ranging from −1 to nearly 0. Under minimum potentiometer settings, the gain effectively approaches zero, resulting in very small output voltages from each filter branch. Consequently, the voltage entering and exiting the summing amplifier is minimal, which is consistent with the observed results.

C. V_{amp} Results at Maximum Settings

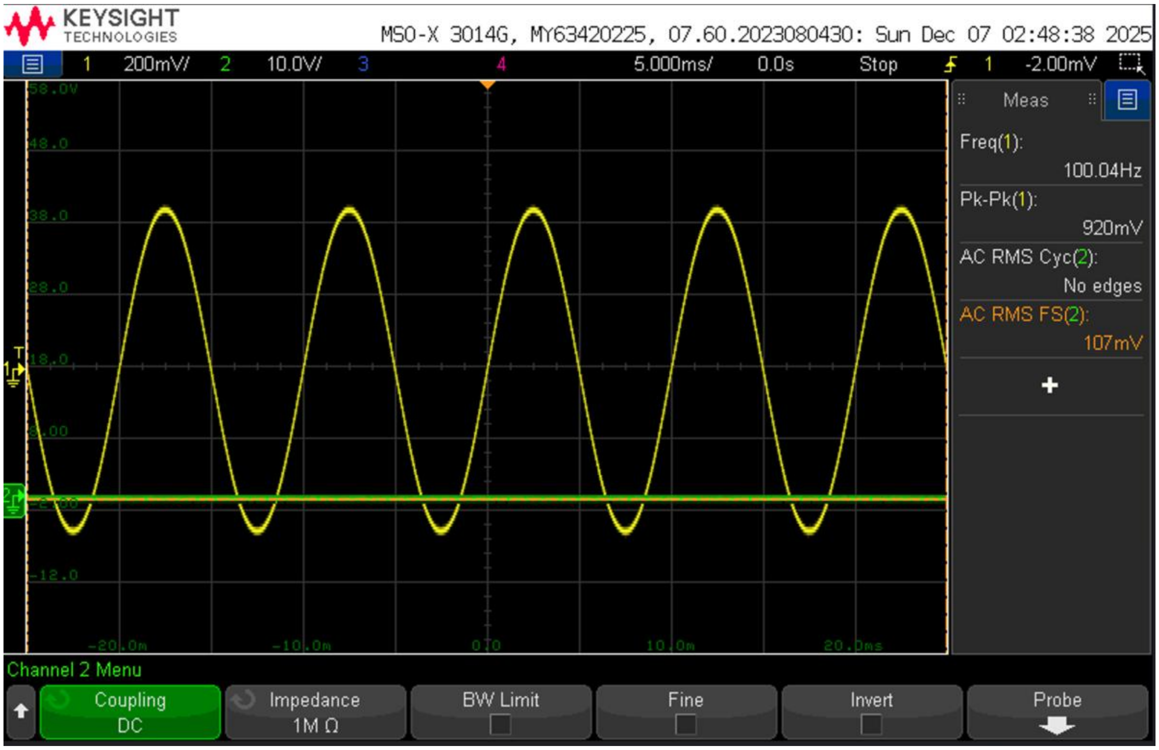

Fig. 15. V_{amp} at minimum settings for 100 Hz

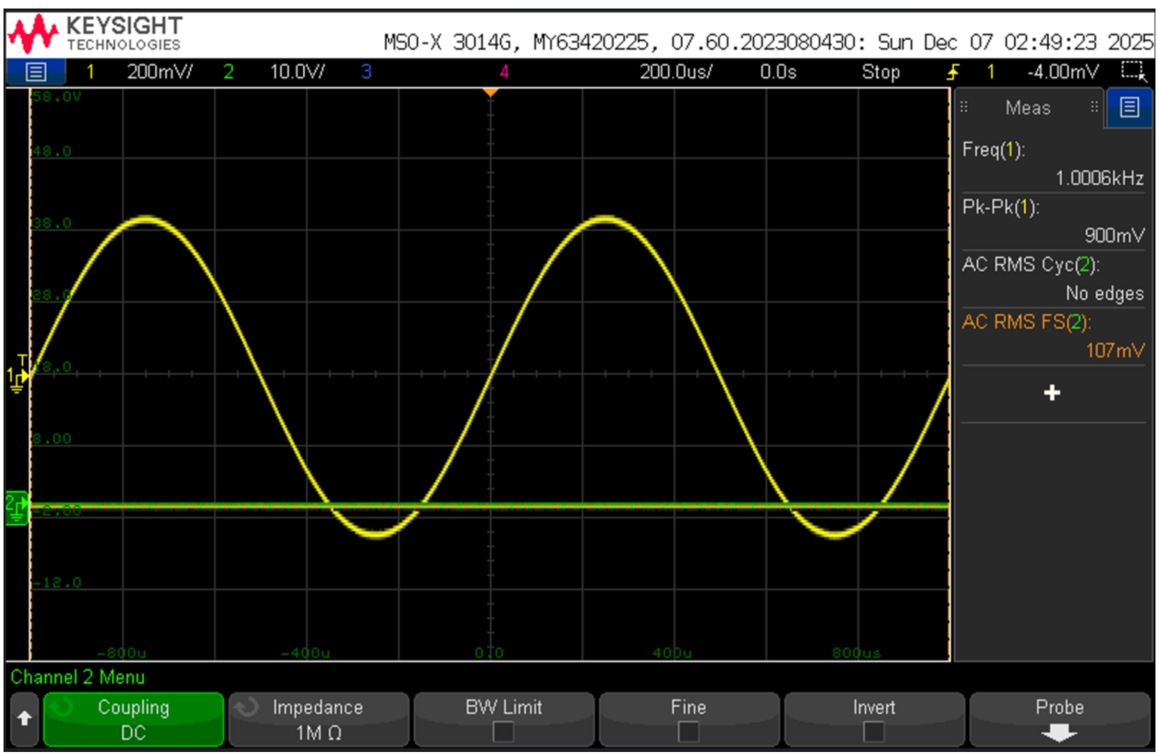

Fig. 16. V_{amp} at minimum settings for 1 kHz

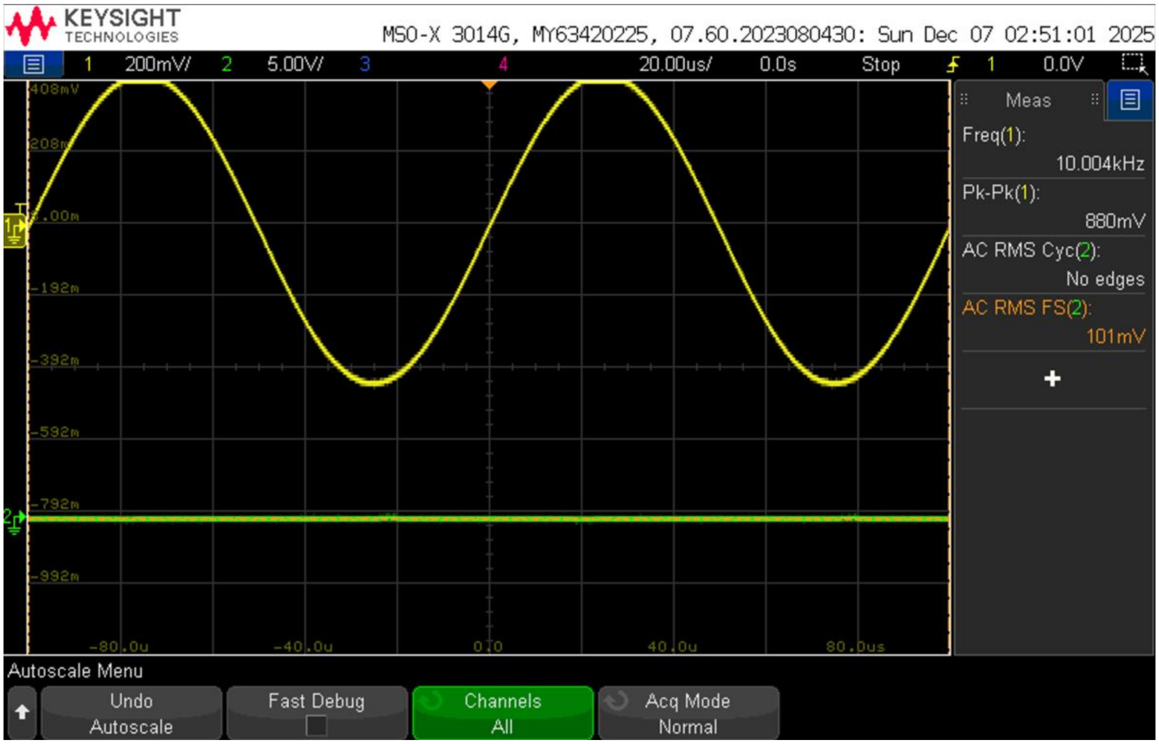

Fig. 17. V_{amp} at minimum settings for 10 kHz

The previous graphs can be summarized as follows: at an input frequency of 100 Hz, the measured output voltage of the summing amplifier was approximately 107 mV_{RMS}. At 1 kHz, the output voltage was also approximately 107 mV_{RMS}, while at 10 kHz the measured output voltage was approximately 101 mV_{RMS}. These results are close to the expected value of 100 mV_{RMS} across the tested frequency range, indicating that the summing amplifier performs consistently under maximum volume settings.

It can be observed that the measured output voltage at each input frequency is approximately 100 mV_{RMS}, which aligns with the expected behavior of the system. As discussed in the theory and calculations section, the selected resistor values for the volume control stages in each branch and for the summing amplifier were chosen such that the gain of each stage ranges from −1 to 0. With all potentiometers set to their maximum values, the signal is allowed to pass through a single frequency branch while being effectively attenuated in the other two. The signal is then inverted by the volume control stage and subsequently inverted again by the summing amplifier, resulting in a properly scaled output voltage of approximately 100 mV_{RMS}.

D. Ripple Results

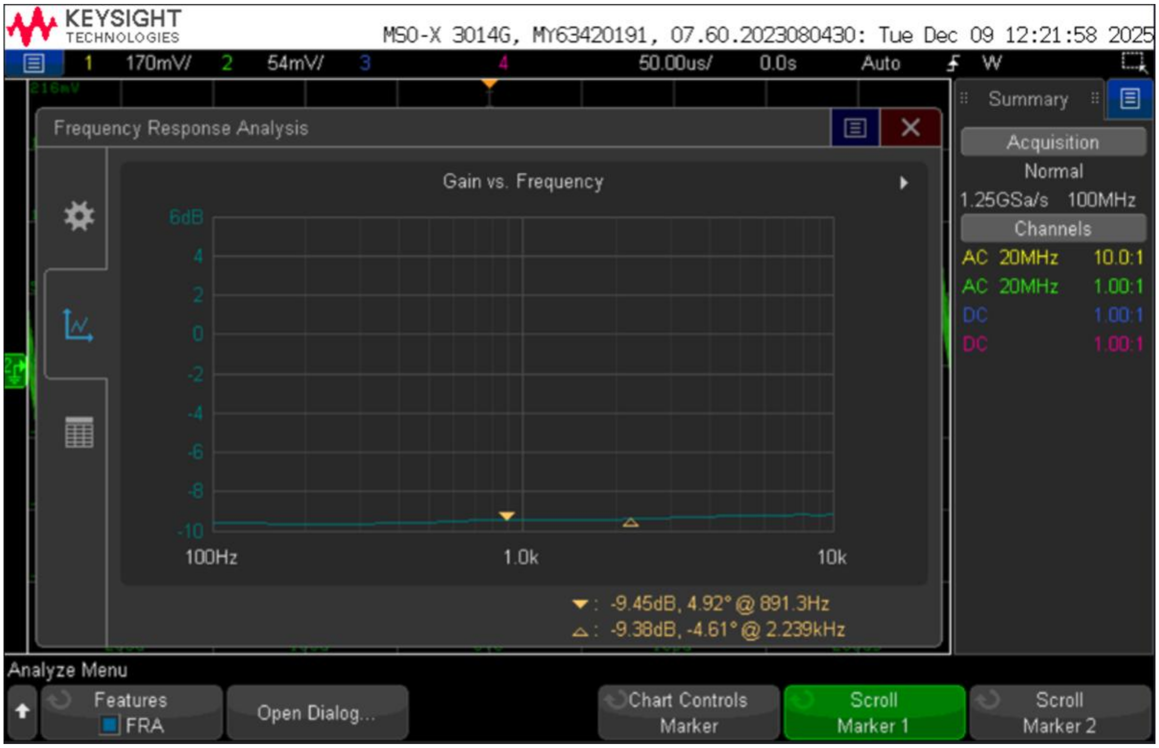

Fig. 18. FRA on the output of the summing amp

The frequency response analysis shown above presents the output of the summing amplifier relative to the input voltage over a frequency range of 100 Hz to 10 kHz. As indicated by the plot, the gain remains relatively constant across the entire frequency spectrum. This consistent gain demonstrates that the ratio of the output voltage to the input voltage is largely independent of frequency, confirming stable summing amplifier performance over the tested range.

Using the two markers shown on the plot (approximately −9.45 dB at 891 Hz and −9.38 dB at 2.239 kHz), the peak-to-peak ripple over that region is

$$\Delta G_{pp} = (-9.38) - (-9.45) = 0.07dB$$

A ripple of 0.07 dB is very small. Converting that to a linear magnitude variation corresponds to an amplitude variation of only approximately 0.8%. In other words, the magnitude of an arbitrary transfer function $| H(f) |$ remains nearly constant across the measured frequency range. This behavior is desirable for the summing amplifier and master volume stage, as it confirms that the circuit does not introduce any unintended frequency selectivity and instead provides uniform gain across the audio band.

Mathematically, this relationship demonstrates that the summing amplifier ideally provides a frequency-independent gain determined solely by the ratio of the feedback resistance to the input resistance. Under the assumption that the input resistors are equal and that only one filter branch is active at any given time, the gain reduces to a constant scale factor. As a result, the summing amplifier does not introduce additional frequency shaping and instead preserves the amplitude relationship established by the preceding filter and volume control stages.

Therefore, any small variations observed in the measured frequency response are not indicative of intentional filtering behavior, but rather arise from practical non-idealities in the circuit. These effects include component tolerances, minor overlap between filter bands near their cutoff frequencies, potentiometer wiper resistance, finite operational amplifier bandwidth, coupling and decoupling capacitor impedance, and limitations of the measurement setup such as probe loading, noise floor, and FRA resolution. The measured ripple of approximately 0.07 dB is sufficiently small to confirm that the summing amplifier maintains a stable and nearly frequency-independent gain across the tested frequency range. This behavior aligns with the intended design objective of the summing and master volume stage and validates its proper operation within the overall audio equalizer system.

E. Output Power

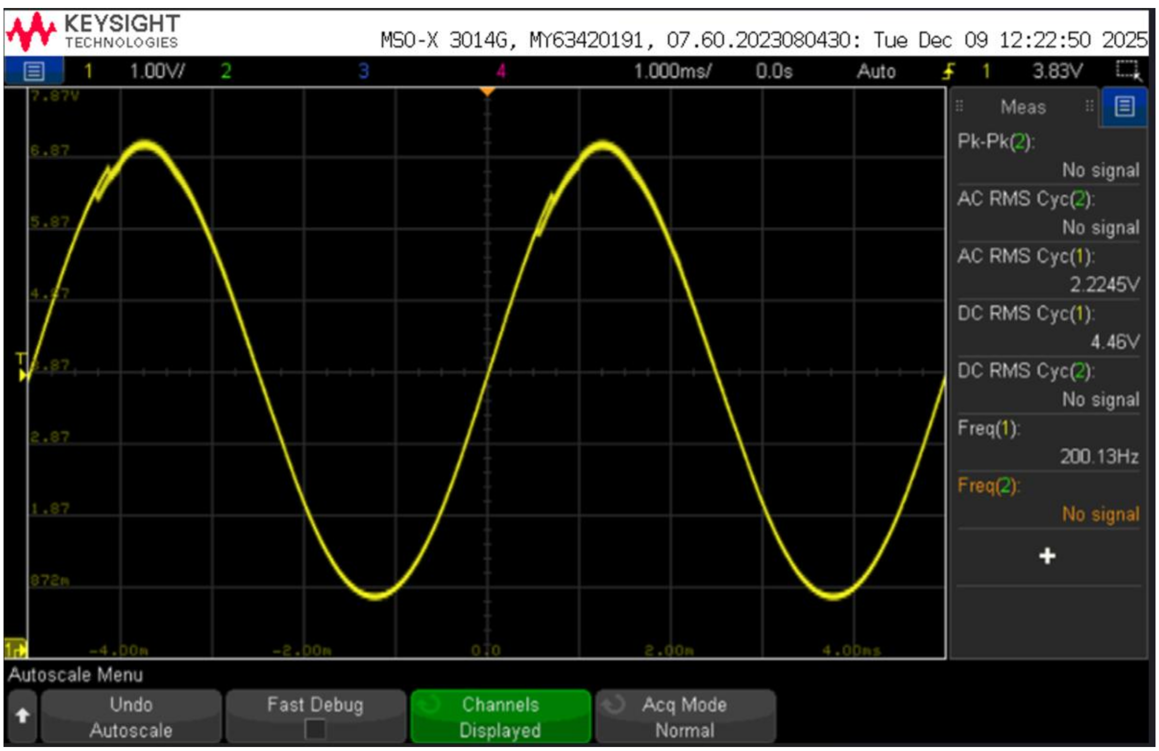

Fig. 19. Ouput voltage at 200 Hz input

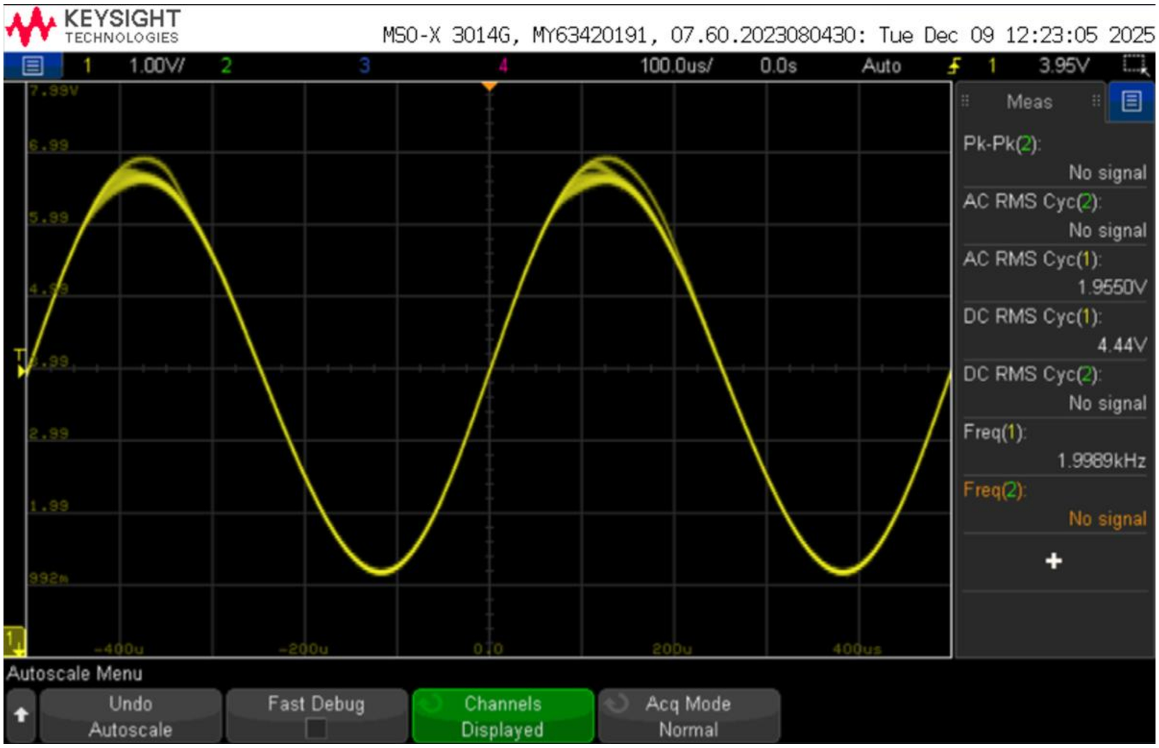

Fig. 20. Output voltage at 2 kHz input

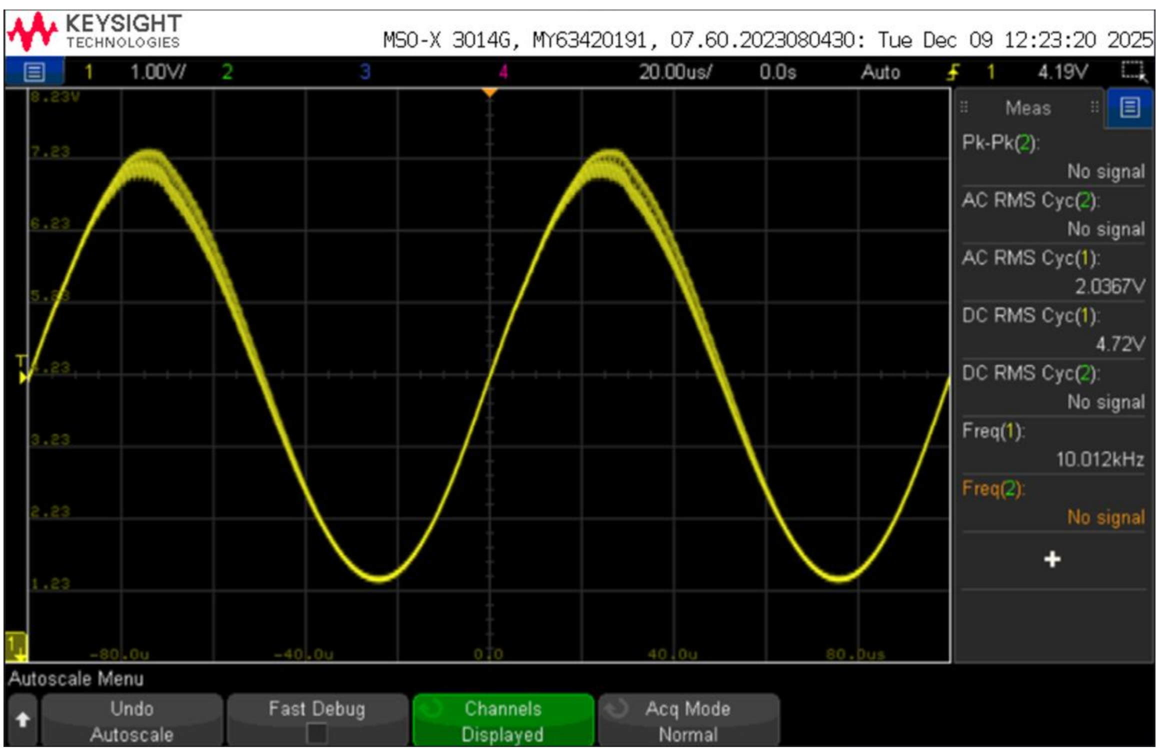

Fig. 21. Output voltage at 10 kHz input

Figures 19, 20, and 21 illustrate the output signal of the complete circuit when a 320 mV_{RSM} input voltage is applied at frequencies of 200 Hz, 1 kHz, and 10 kHz, respectively. At an input frequency of 200 Hz, the measured output voltage was approximately 2.22 V_{RMS}. At 1 kHz, the output voltage was approximately 1.95 V_{RMS}, while at 10 kHz the output voltage was approximately 2.0367 V_{RMS} Using these measured values, Equation (7) can be applied to calculate the output power. The speaker impedance is given as 8 Ω, which is substituted for R in the calculation.

$$P = IV = \frac{V^2}{R} \quad (7)$$

$$P_{200} = \frac{(2.22)^2}{8} = 616.1{\sim}\mathrm{mW}$$

$$P_{2000} = \frac{(1.95)^2}{8} = 475.3{\sim}\mathrm{mW}$$

$$P_{10000} = \frac{(2.0367)^2}{8} = 518.5{\sim}\mathrm{mW}$$

Substituting the measured voltage and resistance values into the power equation yields the output power at each test frequency. At 200 Hz, the output power was approximately 616.1 mW. At 1 kHz, the calculated output power was approximately 475.3 mW, and at 10 kHz the output power was approximately 518.5 mW. All of these values exceed the minimum required output power threshold of 400 mW, confirming that the circuit delivers sufficient power to drive the speaker across the tested frequency range.

F. Low, Medium, and High Volume Control

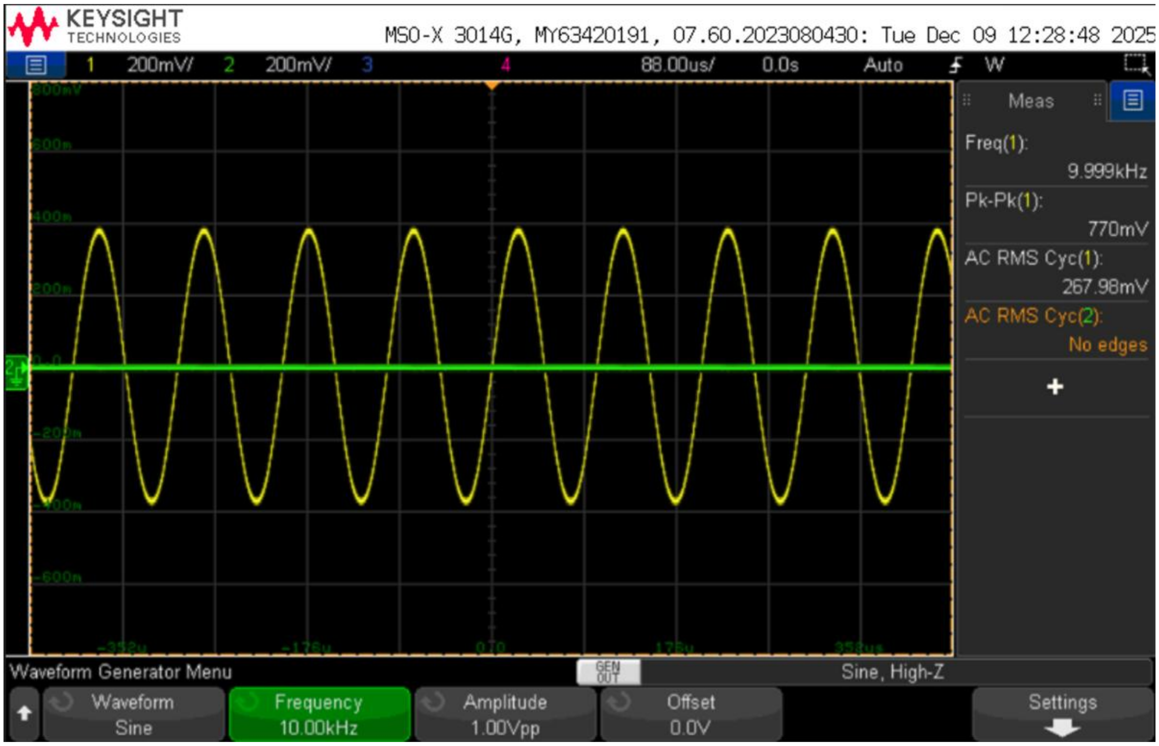

Fig. 22. Low volume control with audio signal input

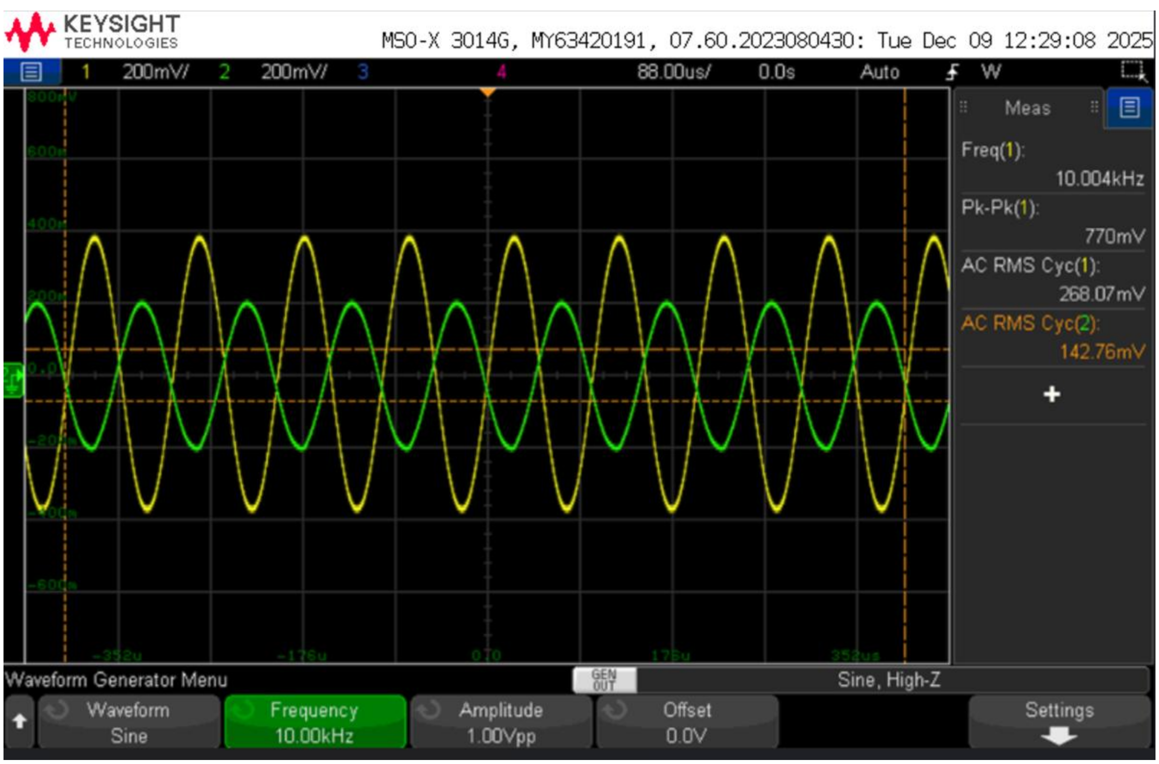

Fig. 23. Medium volume control with audio signal input

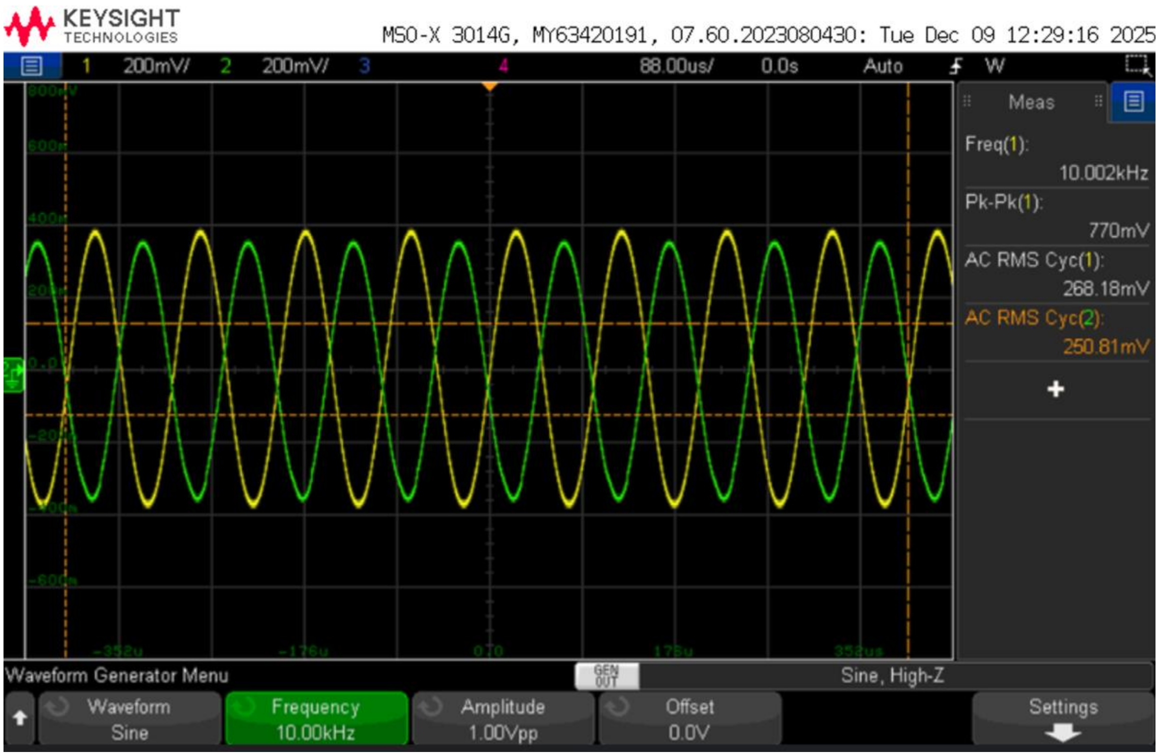

Fig. 24. High volume control with audio signal input

Figures 22, 23, and 24 illustrate the operation of the master volume control when an audio signal is applied to the circuit. In Figure 22, the volume control potentiometer is set to a low position, resulting in a significantly reduced output signal amplitude. Correspondingly, the speaker output was observed to be very quiet at this setting. Figure 23 shows the output signal with the volume control adjusted to a medium level. In this case, the signal amplitude increases relative to the low-volume setting but does not yet reach its maximum value, producing a moderate increase in perceived loudness from the speaker. Figure 24 demonstrates the circuit output when the volume control is set to its maximum position. At this setting, the output signal amplitude is the largest, and the speaker produces its highest audible volume.

Together, these plots demonstrate the effectiveness of the potentiometer within the summing amplifier as a master volume control. By varying the feedback resistance, the potentiometer directly adjusts the gain of the summing amplifier, thereby controlling the amplitude of the output signal and, consequently, the loudness of the speaker. This behavior confirms that the volume control operates as intended and provides smooth, continuous adjustment of the audio output level.

G. Result Justification

Overall, the experimental results closely align with the predicted behavior of the circuit, and all measured parameters fall within the specified threshold ranges. The experimentally determined cutoff frequencies for the low-pass, band-pass, and high-pass filters were all in close agreement with their theoretical values. For the low-pass filter, the measured cutoff frequency was 306 Hz compared to the expected value of 320 Hz, resulting in a percent error of 4.375%. Similarly, the high-pass filter exhibited a measured cutoff frequency of 3268 Hz compared to the expected 3200 Hz, yielding a percent error of 2.125%. For the band-pass filter, the expected cutoff frequencies were 320 Hz and 3200 Hz, while the measured values were 306 Hz and 3060 Hz, corresponding to percent errors of 4.375% for both cutoff points. All of these errors fall well within the acceptable tolerance threshold of 10%.

These discrepancies are most likely attributable to component limitations and practical implementation constraints. In several cases, exact resistor values were not available, requiring the use of series combinations to approximate the calculated values. Additionally, resistors and capacitors inherently include manufacturing tolerances, meaning their actual values may deviate slightly from their nominal ratings. Because the cutoff frequency depends directly on resistance and capacitance, as shown in Equation (1), small variations in these components can produce noticeable shifts in cutoff frequency. Measurement uncertainty also contributed to error, as the cutoff frequencies on the FRA plots were not captured exactly at the −3 dB point.

The measured output voltage values of V_{amp} also behaved as expected at both minimum and maximum volume settings. At the minimum settings, the measured output voltages of 2.11 mV_{RMS}, 2.10 mV_{RMS}, and 2.08 mV_{RMS} at input frequencies of 100 Hz, 1 kHz, and 10 kHz, respectively, were all well below the specified maximum threshold of 15 mV_{RMS}. Although theoretical calculations predict an output of 0 V under these conditions, the nonzero measured values can be attributed to residual resistance within the potentiometers and other circuit components, which prevents the gain from reaching exactly zero.

At the maximum volume settings, the measured percent errors of 3.92%, 4.28%, and 7.63% at 100 Hz, 1 kHz, and 10 kHz, respectively, also remained within the allowable 10% tolerance. Similarly, ripple and output power measurements confirmed proper circuit operation. The maximum observed ripple in V_{amp} was 9.92 mV_{RMS}, which is below the specified limit of 15 mV_{RMS}. Output power calculations yielded values of 479.9 mW, 483.5 mW, and 511.9 mW at 100 Hz, 1 kHz, and 10 kHz, respectively, all exceeding the minimum required power threshold of 400 mW.

Taken together, these numerical results, along with the frequency response analysis and oscilloscope measurements, demonstrate that the audio equalizer operated as intended.

The circuit met all specified performance thresholds and successfully produced a clean, audible output from the speaker, validating the overall design and implementation.

V. CONCLUSION

Overall, this project successfully demonstrated the design, construction, and experimental validation of an audio equalizer system. Throughout the process, filters, operational amplifiers, power amplifiers, and other fundamental circuit components were utilized to develop a fully functional design. The completed circuit is capable of accepting an audio input signal, separating it into distinct frequency bands, independently adjusting the amplitude of each band, recombining the processed signals, amplifying the resulting output, and delivering the signal to a speaker. Potentiometers were effectively incorporated to allow the user to control both the frequency composition of the output signal and the overall volume level. Frequency response analysis and output testing confirmed that the system behaved as intended and satisfied all specified performance requirements. When connected to an audio source and speaker, the circuit was able to reproduce music clearly, validating the overall design and functionality.

During the construction and testing phases, most circuit components performed as expected. The largest discrepancy arose in the summing amplifier stage. Initial calculations indicated that 33 kΩ resistors should be used to route each input signal into the summing amplifier. However, experimental testing revealed that this configuration consistently produced V_{amp} values below the target of 100 mV_{RMS}. To address this issue, the resistor values were adjusted empirically. The optimal values were found to be 19.7 kΩ, 25.3 kΩ, and 27.4 kΩ for the low-pass, band-pass, and high-pass branches, respectively. This discrepancy likely resulted from the simplifying assumptions made during the summing amplifier analysis, which treated the operational amplifier and input signal amplitudes as ideal. In practice, the effective sum of the input voltages is not exactly 320 mV_{RMS}, requiring resistor adjustments to ensure a consistent final output voltage of approximately 100 mV_{RMS} across all frequencies.

Aside from this adjustment, remaining sources of error can largely be attributed to component approximations and tolerances. Exact resistor and capacitor values were not always available, necessitating the use of approximate combinations. Furthermore, real-world components inherently include manufacturing tolerances that can cause deviations from nominal values. Despite these factors, all measured parameters remained within acceptable tolerance limits, and the circuit operated reliably. These results demonstrate both the robustness of the design and its practical applicability to real-world audio signal processing applications.

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